

### 2.1.1 Cells and microscopy

| Learning outcomes                                                                                                  | Additional guidance                                                                                                                                                                                                 |
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| <i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>                      |                                                                                                                                                                                                                     |
| (a) (i) the importance of microscopy in the development of the cell theory and the investigation of cell structure | To include the use of the light microscope, transmission and scanning electron microscopes and recent developments such as the confocal scanning microscope.                                                        |
| (ii) the preparation of blood smears (films) for use in light microscopy                                           | To include how blood smears are made and the interpretation of stained material. Practical work to be carried out in accordance with current CLEAPSS® guidelines. Also see Section 5f.<br><b>PAG1</b><br>HSW4, HSW6 |
| (b) the procedure for differential staining                                                                        | To include the use of Leishman's stain to identify leucocytes in blood smears.<br>HSW4, HSW6                                                                                                                        |

- (c) (i) the structure of animal cells as illustrated by a range of blood cells and components as revealed by the light microscope
- To include red blood cells (erythrocytes), platelets, neutrophils, lymphocytes and monocytes as specialised cells with particular functions related to their structures.
- (ii) the observation, drawing and annotation of cells in a blood smear as observed using the light microscope
- PAG1**
- (d) the linear dimension of cells and the use and manipulation of the magnification formula
- M1.8*  
**PAG1**
- $$\text{magnification} = \frac{\text{image size}}{\text{actual size (of object)}}$$
- (e) practical investigations using a haemocytometer to determine cell counts
- To determine the mean numbers of erythrocytes and convert to a concentration (to include details of dilutions and calculations).  
*M0.1, M0.2, M0.3, M1.5, M1.10, M4.1*  
**PAG1**  
HSW3, HSW4, HSW5, HSW6
- (f) the principles and use of flow cytometry in blood analysis
- To include the use of fluorescent labels. Details of the use of different lasers is not required.  
HSW3, HSW8
- (g) the ultrastructure of a typical eukaryotic animal cell, such as a leucocyte, as revealed by an electron microscope
- To include the structure and function of the following: cell surface membrane, Golgi apparatus, rough endoplasmic reticulum (RER) and smooth endoplasmic reticulum (SER), ribosomes, lysosomes, vesicles, mitochondria, cytoskeleton, centrioles, nucleus and nucleolus.  
HSW8  
*M1.8*
- (h) (i) the ultrastructure of a typical eukaryotic plant cell such as a palisade mesophyll cell and a prokaryotic cell, as revealed by an electron microscope
- To include chloroplasts, large vacuole and the cell wall in plant cells, circular DNA, plasmids, mesosome, pili and flagella in prokaryotic cells.  
HSW8
- (ii) the similarities and differences between the structure of eukaryotic plant and animal cells, and between eukaryotic and prokaryotic cells

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| <p>(i) practical investigations using a graticule and stage micrometer to calculate and measure linear dimensions of cells</p>                           | <p>To include the calibration of an eyepiece graticule using a stage micrometer, calculating the area of the field of view and measuring the sizes of organs, tissues, cells and organelles and calculating their magnification.<br/> <i>M0.1, M0.2, M1.1, M1.2, M1.8, M2.2, M4.1</i><br/> <b>PAG1</b><br/>           HSW3, HSW4, HSW8</p> |
| <p>(j) how the plasma membrane is composed of modified lipids and how the structure of triglycerides and phospholipids is related to their functions</p> | <p>To include reference to fatty acids, glycerol, phosphate groups, ester bonds and hydrophobic/hydrophilic properties.</p>                                                                                                                                                                                                                |
| <p>(k) the fluid mosaic model of the typical plasma membrane</p>                                                                                         | <p>To include the location and function of phospholipids, intrinsic proteins, extrinsic proteins, cholesterol, glycolipids and glycoproteins.</p>                                                                                                                                                                                          |
| <p>(l) the movement of molecules across plasma membranes</p>                                                                                             | <p>To include diffusion and facilitated diffusion as passive methods of transport across membranes<br/> <b>AND</b><br/>           active transport, endocytosis and exocytosis as processes requiring ATP as an immediate source of energy.<br/>           HSW8</p>                                                                        |
| <p>(m) practical investigation(s) into factors affecting diffusion rates in cells</p>                                                                    | <p>To include the use of model cells and tissues such as beetroot.<br/> <i>M0.1, M0.2, M1.1, M1.2, M3.1, M3.2, M3.3, M3.5, M3.6</i><br/> <b>PAG8</b><br/>           HSW1, HSW2, HSW3, HSW4, HSW5, HSW6</p>                                                                                                                                 |
| <p>(n) the roles of membranes within and at the surface of cells</p>                                                                                     |                                                                                                                                                                                                                                                                                                                                            |
| <p>(o) the interrelationship between the organelles involved in the production and secretion of proteins.</p>                                            | <p>To include the role of the cytoskeleton and motor proteins, the nucleus, ribosomes, RER, Golgi and vesicles (no details of transcription and translation are required at this stage).<br/>           HSW8</p>                                                                                                                           |